

CASSANDRA

R3v 1.0.0

User Manual

Interpretable prognostics with Rul3volution

This manual describes the public Cassandra R3v 1.0.0 release prepared for reproducible research use and future alignment with the associated scientific work. It covers installation, the desktop and command-line workflows, experiment artifacts, evaluation, diagnostics, and responsible interpretation of results.

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GNU GPL v3 or later

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1 Purpose and scope

Cassandra R3v is the first public software implementation of the Rul3volution method for Remaining Useful Life (RUL) estimation. Version 1.0.0 operates on the four NASA C-MAPSS turbofan subsets and implements a transparent normalized zero-order Takagi-Sugeno model with readable fuzzy rules.

The software separates four quantities for every rule:

- antecedent activation, denoted by μ_i ;
- consequent, denoted by c_i ;
- positive global relative authority, denoted by a_i ;
- normalized local responsibility, denoted by β_i .

AutoStructure can propose and validate new conjunctions built with AND. Validation is performed on TRAIN engines only. Official TEST endpoints are used after the structure is frozen.

Release identity. Scientific results attributed to Cassandra must be generated from Git tag v1.0.0, with the commit identifier and experiment profile recorded. This release is a public software reference for R3v and is intended to support reproducibility and alignment with the associated article.

1.1 What this release demonstrates

Version 1.0.0 demonstrates that the complete Rul3volution workflow is executable, auditable, and reproducible on a realistic prognostics benchmark. It does not claim state-of-the-art superiority over unrestricted black-box models, causal explanations, or industrial deployment validation.

2 Requirements and installation

2.1 System requirements

- Python 3.10 or newer;
- NumPy 1.24 or newer;
- pandas 2.0 or newer;
- Tkinter for the desktop interface;
- sufficient storage for NASA C-MAPSS and generated artifacts.

No GPU is required. Cassandra can run on x86-64 and ARM64 systems when compatible Python packages are available.

On Debian, Ubuntu, or Linux Mint, install the system components with:

```
sudo apt install python3 python3-venv python3-tk
```

2.2 Install from the source release

From the repository root:

```
python3 -m venv .venv
source .venv/bin/activate
python -m pip install --upgrade pip
python -m pip install .
cassandra-r3v --version
```

The last command must report `cassandra-r3v 1.0.0`.

On Windows, use:

```
py -3 -m venv .venv
.venv\Scripts\activate
python -m pip install .
```

2.3 Start Cassandra

From an unpacked source release:

```
./run_cassandra.sh
```

Alternatives are:

```
python3 run_cassandra.py
cassandra-r3v gui
```

On Windows, double-click `run_cassandra.bat`. The desktop launchers print the Cassandra banner before the window opens. Headless commands do not print the banner.

3 NASA data and the Cassandra workspace

NASA C-MAPSS data are not included in the repository. Obtain the Turbofan Engine Degradation Simulation data from an official NASA source under its applicable terms. Cassandra accepts either the outer NASA archive or the inner `CMAPSSData.zip` archive.

The release acceptance run used an archive with SHA-256:

```
c9c5dec12a945a82e8bb4446589d7fb3cc057b5e5d81fa1a12e25ee9912ad3b2
```

This digest identifies the tested archive. Another archive may contain the same NASA files but have a different digest because of repackaging.

Cassandra stores data and generated evidence in a separate workspace. The default command-line workspace is `cassandra_workspace`. Do not commit this directory to the public repository.

Data boundary. Preparation, standardization, fuzzy partitions, pseudo-endpoints, candidate screening, and structural validation use TRAIN only. TEST data are reserved for final evaluation.

4 Desktop interface

The top bar defines a shared scope. Select one or more subsets from FD001 to FD004 and one or both protocols. The same selections apply to every stage. Cassandra prevents an empty subset or protocol selection.

The workflow strip summarizes prerequisites:

Cassandra R3v 1.0.0 · Interpretable prognostics

File Help

CASSANDRA

Rul3volution · Rule, Evolution, Stability · v1.0.0

Stop processing

Shared scope ⓘ ☒ FD001 ☒ FD002 ☒ FD003 ☒ FD004 | ☒ Full RUL (uncapped) ☒ RUL cap125 8 separate experiments

1 Data 2 Preparation 3 Model 4 AutoStructure 5 Evaluation 6 Full Test

1 · Data 2 · Preparation 3 · Model 4 · AutoStructure 5 · Evaluation 6 · Full Test

Evidence-controlled structural growth

Candidates are compound linguistic concepts. Support and novelty screen them; only matched engine-group validation can accept them.

✓ a parametric model is available for every experiment

Thresholds

Minimum support ⓘ

Minimum novelty ⓘ

NASA improvement ⓘ

Maximum order ⓘ

Composition threshold ⓘ

Pruning threshold (proposal) ⓘ

Computational guard

Maximum time (min) ⓘ

Maximum candidates ⓘ

Screening top-k ⓘ

Validations per seed ⓘ

Maximum rounds ⓘ

Comparison epochs ⓘ

Pre-run estimate ⓘ up to 6,900 pairs · about 45 matched fits per round

Run AutoStructure

Subset	Protocol	Structure
FD001	uncapped	✓ 120 rules · 0 added
FD001	cap125	✓ 120 rules · 0 added
FD002	uncapped	✓ 120 rules · 0 added
FD002	cap125	✓ 120 rules · 0 added
FD003	uncapped	✓ 120 rules · 0 added
FD003	cap125	✓ 121 rules · 1 added
FD004	uncapped	✓ 120 rules · 0 added
FD004	cap125	✓ 120 rules · 0 added

Figure 1: The AutoStructure stage with all eight experiments selected. The table shows seven unchanged backbones and one accepted conjunction for FD003/cap125 in the canonical acceptance campaign.

- **green**: all selected experiments are ready;
- **yellow**: only part of the selected scope is ready;
- **light gray**: a required stage is missing;
- **blue gray**: an optional stage is available;
- Full Test becomes green after a passing report.

Labels marked with an information symbol are clickable. They open a local explanation beside the relevant control. During a calculation, **Stop processing** requests cancellation at the next safe checkpoint. The application remains open and incomplete temporary output is not accepted as a valid artifact.

5 End-to-end guided workflow

Run the stages in order. Each subset/protocol pair is an independent experiment, so selecting all four subsets and both protocols creates eight experiments.

5.1 Stage 1: Data

Choose the NASA archive and import it. Cassandra extracts and verifies the files required for FD001–FD004. Importing does not create prepared experiments.

Successful outcome: the Data row for every available subset is ready.

5.2 Stage 2: Preparation

Run Preparation for the shared scope. Cassandra constructs leakage-safe artifacts separately for **uncapped** and **cap125**.

- **uncapped**: the target is the complete observed RUL range;
- **cap125**: the target is clipped at 125 cycles.

Standardization, fuzzy partitions, and pseudo-endpoints are fitted on TRAIN only. The protocols must never be averaged or treated as interchangeable.

Successful outcome: one NPZ preparation exists for every selected pair.

5.3 Stage 3: Model

Run Model to train the 120-rule Full24 unary backbone. Cassandra uses five linguistic levels for each of the 24 physical channels. Cycle and RUL are not antecedent inputs.

Training writes:

- **model_backbone.json**: the retained frozen reference;
- **model.json**: the current model, initially identical to the backbone.

Successful outcome: both model files exist and the interface reports 120 rules for the initial structure.

5.4 Stage 4: AutoStructure

AutoStructure screens compound linguistic concepts and accepts a conjunction only when support, novelty, matched engine-group validation, and seed consensus satisfy the configured criteria. At most one rule is accepted per round.

The canonical settings include three validation seeds, a two-seed consensus, a maximum order of 3, three rounds, and a 60-minute guard per experiment. The pre-run estimate explains the maximum screening workload.

Returning 120 rules - 0 added is a valid result. It means that no candidate met the evidence requirements. AutoStructure is not required to add a rule in every experiment.

By default, a new AutoStructure run restarts from `model_backbone.json`. Use continuation from the current model only when an explicitly continued structural search is intended.

The pruning threshold produces proposals only. Version 1.0.0 never deletes a rule automatically and does not implement OR consolidation.

5.5 Stage 5: Evaluation

Evaluation applies the frozen backbone and the final model to the same official TEST engines. It writes final-model metrics, local rule decompositions, and a paired backbone–AutoStructure comparison.

Inspect the NASA score first: lower is better. Also inspect MAE, RMSE, R^2 , bias, individual engine penalties, and the uncertainty interval.

For a selected endpoint, the explanation table reports μ_i , a_i , β_i , c_i , and $\beta_i c_i$. The sum of rule contributions reconstructs the prediction exactly. This is a faithful decomposition, not a causal claim.

5.6 Stage 6: Full Test

Full Test performs a non-destructive release and workspace audit. It checks the runtime, source integrity, numerical contracts, source tests, raw data, preparations, models, AutoStructure reports, evaluations, paired comparisons, short pipeline paths, and a bounded structural-validation path.

A TXT report is saved automatically under `cassandra_workspace/reports`. Use **Save Report As...** to copy it elsewhere.

Meaning of PASS. A Full Test PASS proves that the calculation and artifacts are valid. It does not prove that AutoStructure improved predictive performance.

6 Worked example: FD003 with cap125

This example illustrates how to read an AutoStructure outcome. Select only FD003 and RUL `cap125`, then run all six stages in order with the canonical profile.

In the release acceptance campaign, AutoStructure accepted this conjunction:

`OS1_Altitude IS LOW AND S20_W31_HPTCoolantBleed IS LOW`

The structure grew from 120 to 121 rules. Official TEST evaluation produced:

Quantity	Value
Backbone NASA score	855.452
Final NASA score	823.922
Point improvement	+3.686%
Paired 95% bootstrap interval	[-7.676%, +9.488%]
Conclusion	inconclusive

The correct interpretation is precise:

- AutoStructure executed correctly and accepted one TRAIN-validated rule;
- the final TEST point estimate was better;
- the 95% interval included zero;
- therefore, the TEST improvement was not statistically supported at the declared interval level.

Small floating-point differences can occur across platforms. Structural decisions near a threshold must be checked against saved artifacts rather than assumed to be bitwise identical.

7 Interpreting the paired comparison

The paired analysis resamples complete TEST engines 20,000 times with the fixed seed 20260916. Positive gain means that the final NASA score is lower.

Conclusion	Meaning
<code>no_change</code>	Backbone and final endpoint predictions are identical.
<code>supported_improvement</code>	The complete 95% gain interval is above zero.
<code>supported_degradation</code>	The complete 95% gain interval is below zero.
<code>inconclusive</code>	The interval includes zero. The point estimate alone is not enough to claim improvement or degradation.

TEST results never accept, reject, or tune a rule. Their purpose is final measurement after TRAIN-only structural selection.

8 Command-line workflow

The desktop and command-line interfaces use the same implementation and artifact contract. The complete canonical sequence is:

```
cassandra-r3v --workspace cassandra_workspace \  
  import-data /path/to/nasa_cmapss.zip  
  
cassandra-r3v --workspace cassandra_workspace prepare \  
  --subsets FD001 FD002 FD003 FD004 \  
  --protocols uncapped cap125  
  
cassandra-r3v --workspace cassandra_workspace train \  
  --subsets FD001 FD002 FD003 FD004 \  
  --protocols uncapped cap125 \  
  --epochs 250 --learning-rate 0.025 \  
  --ridge 0.9 --patience 50  
  
cassandra-r3v --workspace cassandra_workspace autostructure \  
  --subsets FD001 FD002 FD003 FD004 \  
  --protocols uncapped cap125 \  
  --min-support 0.01 \  
  --activation-support-cutoff 0.01 \  
  --min-novelty 0.005 \  
  --min-relative-improvement 0.01 \  
  --folds 3 --seeds 7 42 99 \  
  --consensus-required 2 --max-order 3 \  
  --screen-top-k 24 \  
  --pruning-threshold 0.002 \  
  --parent-threshold 0.0002 \  
  --validation-epochs 100 --max-minutes 60 \  
  --max-candidates 10000 \  
  --validations-per-seed 4 --max-rounds 3  
  
cassandra-r3v --workspace cassandra_workspace evaluate \  
  --subsets FD001 FD002 FD003 FD004 \  
  --protocols uncapped cap125  
  
cassandra-r3v --workspace cassandra_workspace full-test \  
  --subsets FD001 FD002 FD003 FD004 \  
  --protocols uncapped cap125
```

The exact scientific profile is stored in `examples/article_profile.json`. If an experiment changes these values, the replacement profile and complete command line must be disclosed.

9 Artifacts and evidence

```
cassandra_workspace/
|-- raw/extracted/
|-- prepared/FDxxx/{uncapped,cap125}.npz
|-- models/FDxxx/{uncapped,cap125}/
|   |-- model_backbone.json
|   '-- model.json
|-- structures/FDxxx/{uncapped,cap125}/autostructure.json
|-- evaluations/FDxxx/
|   |-- {uncapped,cap125}.json
|   '-- {uncapped,cap125}_backbone_vs_autostructure.json
|-- reports/cassandra_full_test_<timestamp>.txt
'-- archive/<timestamp>/
```

Artifact	Purpose
prepared/*.npz	TRAIN-fitted arrays, targets, partitions, and metadata.
model_backbone.json	Frozen 120-rule reference model.
model.json	Current final model after any accepted growth.
autostructure.json	Candidate screening, validation, consensus, and accepted rules.
evaluation JSON	Final TEST metrics and faithful local decompositions.
*_backbone_vs_autostructure.json	Paired TEST effect and bootstrap uncertainty.
Full Test TXT	Software and artifact acceptance evidence.

Before using a number in an article, preserve the tag, commit SHA, environment, machine description, command line, data hashes, experiment profile, generated artifacts, and Full Test report. Article tables must come from evaluation JSON, not from smoke-test metrics.

10 Troubleshooting

The GUI does not open

Confirm that Tkinter is installed and that a graphical session is available:

```
python3 -c "import tkinter; print(tkinter.TkVersion)"
```

On a headless server, use the command-line workflow.

A stage is disabled

Read the dependency strip and the message at the top of the tab. Complete the earliest missing prerequisite. For example, Model requires Preparation, and Evaluation requires a model.

AutoStructure adds no rules

This is not an error. Open `autostructure.json` or inspect the GUI status to determine whether candidates failed support, novelty, validation, consensus, or a computational guard.

AutoStructure takes too long

Run experiments separately. Screening top-k, candidate, validation, time, and round limits bound the workload. Changing canonical values creates a new experimental profile and must be reported.

Full Test reports SKIP

Missing NASA data or scientific artifacts may be skipped because they are not distributed with the source. A present but invalid artifact is not treated as a skip. For release acceptance, import the data, generate all eight experiments, and rerun Full Test.

A previous artifact becomes outdated

Cassandra archives dependent downstream artifacts when an upstream stage is recomputed. Regenerate the affected stages instead of mixing files from different runs.

11 Scientific limitations and responsible use

Version 1.0.0 has explicit boundaries:

- fuzzy partitions remain fixed after TRAIN fitting;
- instantaneous inputs are used without age or temporal gradients;
- candidate generation can be expensive despite filters and hard guards;
- local decompositions are faithful but not causal;
- pruning proposals do not remove rules automatically;
- disjunctive OR consolidation is not implemented;
- one paired TEST set can leave a structural effect inconclusive;
- no claim of industrial deployment or universal predictive superiority is made.

These limitations do not invalidate Cassandra as an implementation. They define the claims that can be supported by the current release and identify directions for later research.

12 Further documentation and citation

The repository includes:

- README.md: immediate overview and quick start;
- docs/GUI_GUIDE.md: detailed interface behavior;
- docs/REPRODUCIBILITY.md: canonical scientific campaign;
- docs/METHOD_ALIGNMENT.md: method-to-code contract;
- docs/VALIDATION_REPORT.md: release evidence;
- docs/PUBLISHING.md: GitHub publication checklist;
- CITATION.cff: machine-readable citation metadata.

Using and citing Cassandra

If Cassandra is used in scientific work, cite the tagged software release and the associated article. After archival, use the DOI displayed on the release page. The software is distributed under GNU GPL v3 or later.

Recommended reporting sentence. “Cassandra R3v 1.0.0 provides a reproducible first implementation and feasibility demonstration of the Rul3volution workflow on NASA C-MAPSS. Predictive effects of structural growth are reported with paired uncertainty and are not generalized beyond the available evidence.”